

Climate 5000 Series - Gen 4

30K BTU Single Zone Light Commercial System

Wall Mounted IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

30K Btu Single Zone Light Commercial System - Wall Mounted IDU

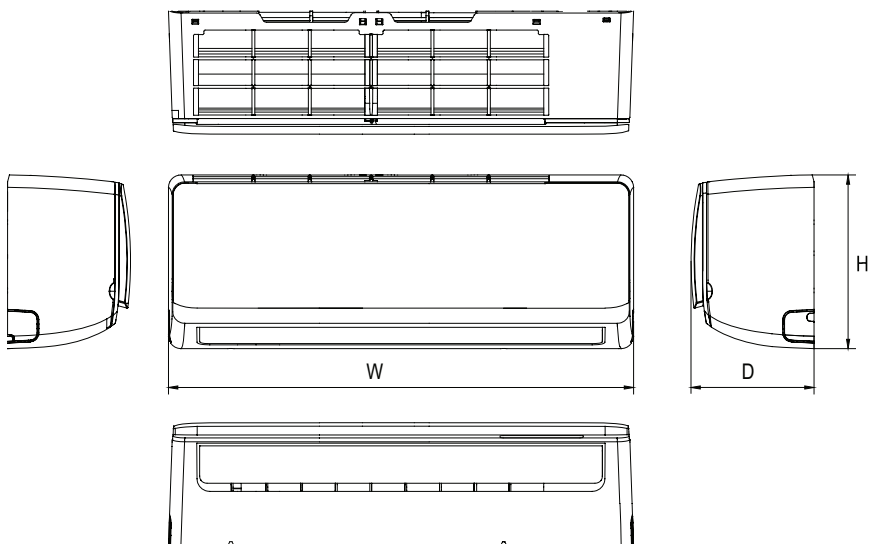
Indoor part number		8-733-962-677
Indoor model number		BMS500-AAU030-1AHWXD
Outdoor part number		8-733-962-678
Outdoor model number		BMS500-AAS030-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		20.5
EER2		12.0
HSPF2 Region IV		9.5
HSPF2 Region V		7.2
COP(47F)		3.08
COP (5F)		2.03
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	6500~32800
	Rated Capacity	30000
	Current	11.2
	Input Power	2500
Heating	Capacity Range	12600~31400
	Rated Capacity (47F)	30000
	Rated Capacity (5F)	20000
	Current	13.1
	Input Power	2857
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 28.0
Max. fuse indoor/outdoor (A)		15.0 / 30.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 670
Input (W)		2200
Rated current (RLA/A)		10.0
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

Piping		
Refrigerant precharge (oz/kg)		84.66 / 2.4
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		3/8" / Ø9.52
Gas side (in/mm)		5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.32 / 30
Max. refrigerant pipe length (ft/m)		164.00 / 50
Max. difference in height (ft/m)		82.00 / 25
System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		8.45
IDU	Fan motor model	ZKFP-58-8-1-10
	Fan motor qty (ea)	1
	Fan motor input (W)	58.0
	Fan speed (RPM)	1050 / 880 / 630
	Air flow, Turbo / H / M / L / Si (CFM)	882.9 / 635.7 / 506.2 / 382.6 / /
	Approx. Air Throw Range ft (m)	29.5 / 9
	Noise level, Turbo / H / M / L / Si (dB(A))	53.0 / 45.5 / 40 / 30.5 / 19.5
	Net weight (lbs / kg)	43.21 / 19.6
	Gross weight (lbs / kg)	54.67 / 24.8
ODU	Fan motor model	ZKFN-120-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	170
	Fan motor speed, H / M / L (RPM)	900 / 700 / 500
	Air flow, Max (CFM)	2235.3
	Noise level, Max (dB(A))	61.0
	Net weight (lbs / kg)	142.86 / 64.8
	Gross weight (lbs / kg)	154.54 / 70.1
Accessories		
BMS-WT2-XXD (8733962707)		Wired Wall Thermostat
BMS-DRY-XXD (8733969263)		24V Interface
BMS-PK3-7X1 (8733951015)		25ft Lineset
Standard Parts		
8-733-969-088		Remote Control

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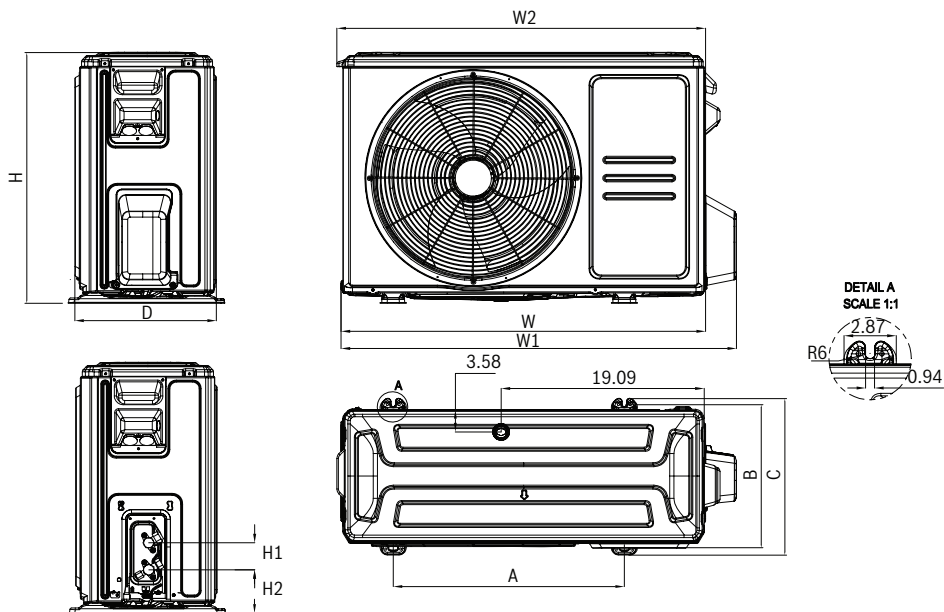


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU030-1AHWXD	44.88	11.22	14.61

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS030-1CSXRD	37.24	16.14	31.89	40.55	37.24	2.36	4.25	26.5	15.87	17.9

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Cooling Performance - Indoor Dry Bulb Temperature at 882 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	22852	22882	22907	22908	22939	24229	24257	24281	24286	24315	25676	25710	25738	25743	25772	27223	27253	27283	27287	27321
	S/T	0.48	0.56	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.77	0.54	0.63	0.72	0.72	0.82	0.57	0.67	0.76	0.76	0.87
	PI	1.16	1.17	1.17	1.17	1.17	1.23	1.24	1.24	1.24	1.24	1.31	1.31	1.31	1.31	1.31	1.39	1.39	1.39	1.39	1.39
-4°F (-20.0°C)	TC	24609	24643	24669	24670	24703	26093	26123	26149	26154	26185	27651	27687	27718	27723	27754	29317	29349	29381	29386	29422
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.82	0.57	0.67	0.77	0.77	0.87
	PI	1.25	1.26	1.26	1.26	1.26	1.33	1.33	1.33	1.33	1.33	1.41	1.41	1.41	1.41	1.41	1.49	1.50	1.50	1.50	1.50
0°F (-17.8°C)	TC	25391	25425	25452	25453	25488	26921	26953	26979	26985	27017	28529	28566	28598	28603	28635	30248	30281	30314	30319	30356
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.29	1.30	1.30	1.30	1.30	1.37	1.37	1.38	1.38	1.38	1.45	1.46	1.46	1.46	1.46	1.54	1.54	1.55	1.55	1.55
5°F (-15.0°C)	TC	26367	26403	26431	26432	26468	27956	27989	28017	28023	28056	29626	29665	29698	29704	29737	31412	31446	31480	31485	31524
	S/T	0.49	0.57	0.65	0.65	0.74	0.51	0.60	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.34	1.35	1.35	1.35	1.35	1.42	1.43	1.43	1.43	1.43	1.51	1.51	1.51	1.51	1.52	1.60	1.60	1.60	1.60	1.61
14°F (-10.0°C)	TC	25795	25829	25857	25858	25893	27349	27382	27409	27414	27447	28983	29021	29053	29059	29091	30729	30763	30796	30801	30840
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.38	1.38	1.38	1.38	1.38	1.46	1.46	1.46	1.46	1.47	1.55	1.55	1.55	1.55	1.55	1.64	1.64	1.65	1.65	1.65
17°F (-8.3°C)	TC	25604	25638	25666	25667	25702	27147	27179	27206	27211	27244	28769	28806	28838	28844	28876	30502	30535	30569	30573	30611
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.79	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.78	0.78	0.88
	PI	1.39	1.39	1.39	1.39	1.39	1.47	1.48	1.48	1.48	1.48	1.56	1.56	1.56	1.57	1.57	1.66	1.66	1.66	1.66	1.66
23°F (-5.0°C)	TC	25991	26026	26054	26055	26090	27557	27590	27617	27622	27655	29203	29241	29274	29279	29312	30963	30997	31030	31035	31074
	S/T	0.49	0.58	0.65	0.65	0.75	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.58	0.69	0.78	0.78	0.89
	PI	1.37	1.37	1.37	1.37	1.37	1.45	1.45	1.45	1.45	1.45	1.53	1.54	1.54	1.54	1.54	1.63	1.63	1.63	1.63	1.63
35°F (1.7°C)	TC	26764	26800	26829	26830	26866	28377	28411	28439	28444	28478	30072	30111	30145	30151	30184	31884	31919	31954	31959	31998
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.32	1.32	1.32	1.32	1.32	1.40	1.40	1.40	1.40	1.40	1.48	1.48	1.48	1.48	1.49	1.57	1.57	1.57	1.57	1.58
41°F (5.0°C)	TC	26569	26605	26634	26635	26671	28171	28204	28232	28237	28271	29854	29892	29926	29931	29965	31652	31687	31721	31726	31766
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.35	1.35	1.35	1.35	1.35	1.43	1.43	1.43	1.43	1.44	1.52	1.52	1.52	1.52	1.52	1.61	1.61	1.61	1.61	1.61
47°F (8.3°C)	TC	26375	26410	26438	26440	26475	27964	27997	28025	28030	28064	29635	29673	29706	29712	29745	31420	31454	31489	31494	31533
	S/T	0.50	0.58	0.66	0.66	0.75	0.53	0.62	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.85	0.59	0.69	0.79	0.79	0.90
	PI	1.38	1.38	1.38	1.38	1.38	1.46	1.46	1.47	1.47	1.47	1.55	1.55	1.55	1.55	1.56	1.64	1.65	1.65	1.65	1.65
50°F (10.0°C)	TC	26855	26891	26920	26921	26957	28473	28507	28535	28541	28575	30174	30214	30247	30253	30287	31992	32027	32062	32067	32107
	S/T	0.50	0.58	0.66	0.66	0.76	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90
	PI	1.40	1.40	1.41	1.41	1.41	1.49	1.49	1.49	1.49	1.49	1.58	1.58	1.58	1.58	1.58	1.67	1.67	1.67	1.67	1.68
59°F (15.0°C)	TC	28296	28334	28364	28366	28404	30001	30037	30066	30072	30108	31794	31835	31870	31876	31912	33709	33746	33783	33788	33830
	S/T	0.50	0.59	0.67	0.67	0.76	0.53	0.62	0.71	0.71	0.81	0.56	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91
	PI	1.47	1.47	1.47	1.47	1.48	1.56	1.56	1.56	1.56	1.56	1.65	1.65	1.66	1.66	1.66	1.75	1.75	1.75	1.76	1.76
68°F (20.0°C)	TC	29737	29777	29809	29810	29851	31529	31566	31598	31604	31641	33413	33456	33493	33500	33537	35426	35464	35503	35509	35553
	S/T	0.51	0.60	0.69	0.69	0.78	0.54	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.61	0.72	0.82	0.82	0.93
	PI	1.54	1.54	1.54	1.54	1.54	1.63	1.63	1.63	1.63	1.64	1.73	1.73	1.73	1.73	1.73	1.83	1.83	1.84	1.84	1.84
77°F (25.0°C)	TC	31178	31220	31254	31255	31297	33057	33096	33129	33135	33175	35032	35077	35117	35123	35162	37143	37183	37224	37229	37276
	S/T	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	0.62	0.73	0.83	0.83	0.94
	PI	1.61	1.61	1.61	1.61	1.61	1.70	1.70	1.71	1.71	1.71	1.80	1.81	1.81	1.81	1.81	1.91	1.91	1.92	1.92	1.92
82°F (27.8°C)	TC	31342	31385	31418	31420	31462	33231	33271	33304	33310	33350	35217	35262	35302	35308	35348	37339	37379	37420	37426	37472
	S/T	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	0.63	0.74	0.84	0.84	0.95
	PI	1.79	1.79	1.79	1.79	1.79	1.89	1.90	1.90	1.90	1.90	2.01	2.01	2.01	2.01	2.01	2.13	2.13	2.13	2.13	2.14
86°F (30.0°C)	TC	30671	30713	30746	30747	30789	32520	32558	32591	32597	32636	34463	34507	34546	34552	34591	36539	36579	36619	36624	36670
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	2.11	2.11	2.12	2.12	2.12	2.24	2.24	2.24	2.24	2.24	2.35	2.37	2.37	2.38	2.38	2.51	2.52	2.52	2.52	2.52
95°F (35.0°C)	TC	29162	29201	29232	29233	29273	30919	30956	30986	30992	31029	32766	32809	32845	32851	32888	34740	34778	34816	34821	34865
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	2.84	2.84	2.85	2.85	2.85	3.01	3.02	3.02	3.02	3.02	3.19	3.20	3.20	3.20	3.20	3.38	3.39	3.39	3.39	3.40
104°F (40.0°C)	TC	25639	25674	25701	25702	25737	27184	27216	27243	27249	27281	28808	28846								

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Heating Performance - Indoor Dry Bulb Temperature at 882 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	15365	15057	14982	14922	14788	2.86	2.88	2.88	2.89	2.90
-4°F(-20.0°C)	22044	21603	21495	21409	21216	0.00	0.00	0.00	0.00	0.00
0°F(-17.8°C)	21439	21010	20905	20821	20634	1.30	1.31	1.31	1.31	1.32
5°F(-15.0°C)	20682	20269	20167	20087	19906	2.92	2.95	2.95	2.96	2.97
14°F(-10.0°C)	19321	18935	18840	18765	18596	2.93	2.95	2.96	2.96	2.98
17°F(-8.3°C)	18867	18490	18398	18324	18159	2.93	2.95	2.96	2.97	2.98
22°F(-5.6°C)	19800	19404	19307	19230	19057	2.96	2.98	2.99	2.99	3.01
27°F(-2.8°C)	20733	20319	20217	20136	19955	2.99	3.01	3.02	3.02	3.04
32°F(-0.0°C)	21666	21233	21127	21042	20853	3.02	3.04	3.05	3.05	3.07
35°F(1.7°C)	22226	21782	21673	21586	21392	3.03	3.06	3.07	3.07	3.08
37°F(2.8°C)	23888	23410	23293	23200	22991	3.02	3.05	3.05	3.06	3.07
42°F(5.6°C)	28042	27481	27343	27234	26989	2.99	3.01	3.02	3.03	3.04
44.6°F(7.0°C)	29703	29109	28964	28848	28588	2.98	3.00	3.01	3.01	3.03
47°F(8.3°C)	32195	31552	31394	31268	30987	2.96	2.98	2.99	2.99	3.01
52°F(11.1°C)	31071	30449	30297	30176	29904	2.63	2.65	2.65	2.66	2.67
57°F(13.9°C)	29946	29347	29200	29083	28822	2.30	2.31	2.32	2.32	2.33
59°F(15.0°C)	29496	28906	28761	28646	28388	2.16	2.18	2.19	2.19	2.20
62°F(16.7°C)	28821	28244	28103	27991	27739	1.96	1.98	1.98	1.99	2.00
64.4°F(18.0°C)	28371	27804	27664	27554	27306	1.83	1.85	1.85	1.85	1.86

Bolded values were tested. All other values are simulated and for reference only.